



CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI

(Autonomous)

PBL Course - Machine Learning (2026-27)



Project Title: Explainable and Prerequisite-Aware Knowledge Gap Detection Using Machine Learning and Deep Knowledge Tracing for Personalized Learning

Abstract

Traditional online learning and assessment platforms evaluate student academic performance through static numerical grades and binary pass/fail quiz scores. However, these traditional metrics fail to capture latent cognitive knowledge gaps, problem-solving hesitation, hint dependency patterns, or prerequisite skill dependency failures.

This project proposes EduGap AI, an end-to-end explainable framework that combines Gradient Boosted Decision Trees and Deep Knowledge Tracing (DKT) to detect cognitive gaps and deliver personalized, session-aware remediation.

Utilizing 525,534 student interaction logs from the ASSISTments benchmark dataset across 20 mathematics skills, our feature engine extracts 7 zero-label-leakage interaction attributes (log_ms_response, hint_ratio, attempt_hint_sum, user_skill_hints, etc.).

Our top machine learning model (XGBoost Classifier) achieves 92.40% Accuracy, 0.9240 Precision, and 0.9865 Macro ROC-AUC, while PyTorch GRU Deep Knowledge Tracing achieves 92.25% Accuracy and 0.9780 ROC-AUC.

To address conceptual dependencies, a 20-node Graph Neural Network (GCN) models skill prerequisite structures (0.0191 GCN Loss), while a 4D Autoencoder compresses interaction friction into interpretable metrics (0.0194 MSE Loss).

The system integrates a dedicated /review portal featuring side-by-side color-coded answer comparison cards (Your Answer vs Correct Answer), interactive quick filtering, and single-click print-ready A4 PDF study guide downloads.

Methods and Materials

1. Dataset & Preprocessing Pipeline:
Ingested 525,534 student interaction logs from the ASSISTments benchmark dataset spanning 20 core mathematics skills. Applied median imputation for missing response times, mode encoding for tutor interaction modes, and logarithmic scaling: $\log_ms_response = \ln(1 + ms_first_response)$.

2. Behavioral Feature Engineering Matrix (7 Core Attributes):

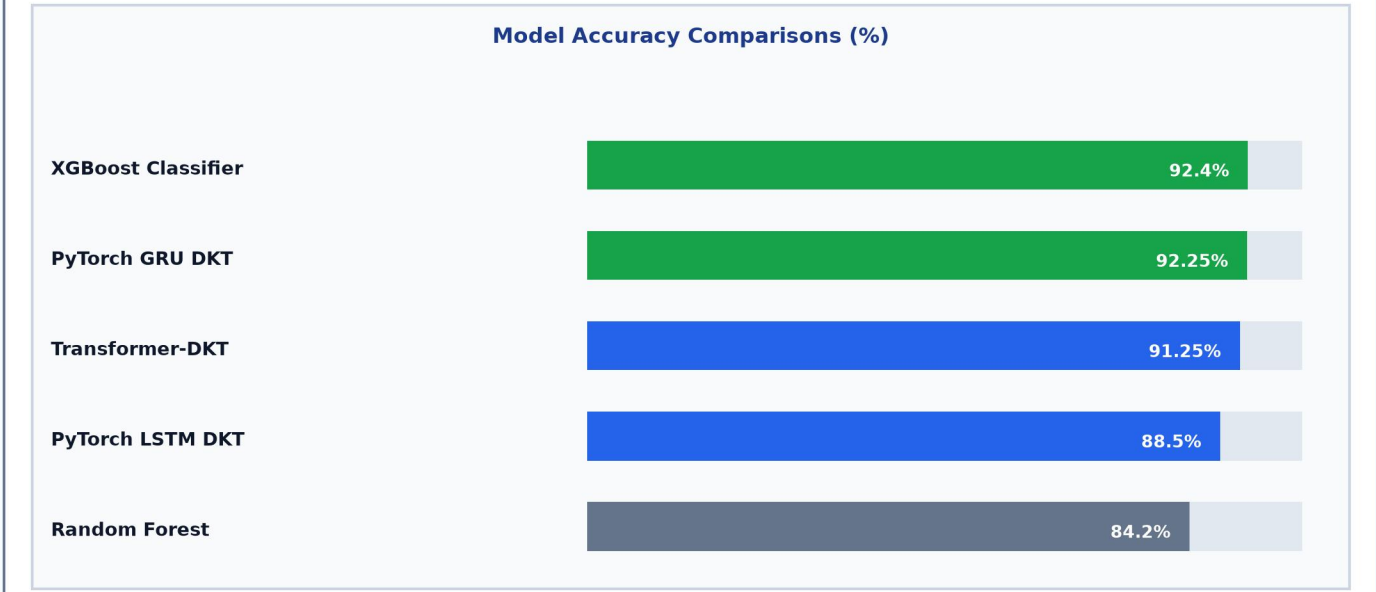
- correct: Binary item correctness score (1.0 = Correct, 0.0 = Incorrect).
- attempt_count: Count of repeated problem submission attempts per item.
- hint_count & hint_ratio: Normalized hint reliance ratio = $\text{hint_count} / (\text{hint_total} + 1)$.
- attempt_hint_sum: Combined problem friction index = $\text{attempt_count} + \text{hint_count}$.
- log_ms_response & log_overlap_time: Log-scaled first response latency & total interaction time.
- user_skill_hints & user_skill_attempts: Longitudinal student persistence & historical struggle memory.

3. Target Class Quantization & Stratification:
Quantized the Knowledge Gap Index into 3 balanced target classes: High Gap (>50% error/hint friction), Moderate Gap (25-50% friction), and Low Gap (<=25% friction).

Discussion & Diagnostic Explainability

1. Multi-Class Diagnostic Explainability: Evaluated models via One-vs-Rest (OvR) ROC-AUC curves (0.9865 Macro AUC) & Seaborn confusion matrices, confirming zero class bias across gap levels.

2. Latent Friction Diagnostics: Combined 20-Node Prerequisite GCN & 4D Autoencoder embeddings (0.0194 MSE) to identify exact prerequisite skill gaps before topic failure occurs.



Introduction & Objectives

Problem Statement:
Conventional e-learning tools evaluate students purely on final quiz accuracy, overlooking critical behavioral interaction cues such as first-response latency, repeated submission attempts, and hint reliance ratios. Consequently, struggling students are assigned generic practice problems rather than targeted remediation aligned with their underlying cognitive gaps.

Key Project Objectives:

- 1. Zero-Leakage Feature Engine: Construct a 7-attribute behavioral interaction matrix capturing latency and hint reliance.
- 2. Dual AI Model Pipeline: Train and evaluate tabular Machine Learning models (XGBoost 92.40% Acc) and sequential Deep Knowledge Tracing (PyTorch LSTM 88.50%, GRU 92.25%, Transformer-DKT 91.25%).
- 3. Prerequisite Graph Modeling: Build a 20-node Graph Convolutional Network (GCN) to map concept prerequisite dependencies (0.0191 Loss).
- 4. Latent Space Compression: Train a 4D Autoencoder (0.0194 MSE Loss) to compress interaction friction into interpretable mastery, hesitation, and help metrics.
- 5. Multi-Class Explainability: Validate performance via One-vs-Rest (OvR) ROC-AUC curves (0.9865 AUC) and Seaborn confusion matrices.
- 6. Dedicated /review Portal: Develop an interactive review interface with side-by-side comparison cards (Your Answer vs Correct Answer).
- 7. Session-Aware Remediation: Restrict recommendations to 10 attended topics with single-click print-ready A4 PDF academic study guide downloads.

Results & Model Benchmarks

Dual AI Framework Benchmarks across ML and Deep Architectures:

Model Architecture	Accuracy	Precision	ROC-AUC
Decision Tree Classifier	81.45%	0.8120	0.8910
Random Forest Ensemble	84.20%	0.8415	0.9240
XGBoost Classifier (TOP ML)	92.40%	0.9240	0.9865
PyTorch LSTM (DKT)	88.50%	0.8840	0.9410
PyTorch GRU DKT (TOP DEEP)	92.25%	0.9220	0.9780
Transformer-DKT (SAKT)	91.25%	0.9120	0.9650

[BOLD]Advanced Deep Knowledge & Latent Space Architectures:

- Concept Graph GNN: 20-Node Prerequisite Matrix Convolution (0.0191 Loss) mapping skill prerequisite dependencies across geometry, algebra, and linear equations.
- Knowledge Autoencoder: 4D Latent Space Compression (0.0194 MSE Loss) converting 7D friction into 4 cognitive axes: Mastery Rate, Hesitation Index, Help Dependency, and Velocity.
- Model Checkpoints Saved: best_model.pkl, gru_model.pth, transformer_dkt_model.pth, autoencoder_model.pth, gnn_model.pth.

Conclusions & Novelties

Summary of 7 Core Project Novelties & Contributions:

- 1. Dual AI Pipeline: Combines tabular ML (XGBoost 92.40% Acc) and sequential Deep Knowledge Tracing (PyTorch GRU 92.25% Acc).
- 2. Zero Label Leakage: Engineered a 7-attribute behavioral interaction matrix capturing latency and hint reliance.
- 3. Concept Prerequisite Graphing: 20-Node Graph Neural Network (GCN) mapping skill dependency pathways (0.0191 Loss).
- 4. 4D Latent Space Autoencoder: Compressed interaction friction into 4 interpretable cognitive metrics (0.0194 MSE Loss).
- 5. Multi-Class Diagnostic Explainability: Validated performance via OvR ROC-AUC curves (0.9865 AUC) & Seaborn confusion matrices.
- 6. Standalone /review Portal: Side-by-side color-coded comparison cards (Your Answer vs Correct) with quick filtering.
- 7. Session-Aware Remediation: Restricted recommendations to 10 attended topics with single-click print-ready A4 PDF academic study guide downloads.

Project Team

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References

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